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Exploring Big Data on a Desktop: Where to Start

Big Data is the current buzz word in the world of computing. Typically, for computer enthusiasts with just a single desktop, experimenting with Big Data poses a problem since running a distributed data program requires many computers. But now, Big Data can be run on a single desktop.

I read about Apache Spark in a news article titled 'Run programs up to 100x faster than Hadoop MapReduce', and I wanted to explore it. Spark applications can be written in Python. The trouble was that I hadn't even got down to trying Hadoop MapReduce, in spite of wanting to do so for years.

If all you have is a desktop, how do you experiment and learn systems that are inherently distributed? You can't get an insight into them while using a single machine.

Fortunately, even mid-range desktops now come with a quad-core and 8 GB of RAM. So, you can run at least a few virtual machines. Many tools, e.g., VirtualBox, are very good and easy to use. But are they enough?

The scope

Spark is a "...fast and general engine for large-scale data processing" (<http://spark.apache.org>). It allows you to build parallel applications. It can access the data from various sources, particularly, existing Hadoop data. It can run on the YARN cluster manager of Hadoop 2. So, you may want to understand and set up a Hadoop data source and, possibly, a YARN cluster manager.

You need to set up a cluster of virtual machines on which the master and slave instances of Spark will run. Then set up a HDFS cluster of virtual machines for Hadoop data. There will be a NameNode to manage the file system metadata and DataNodes that will store the actual data.

You may need to play around with the number of virtual machines in order to avoid manually creating each virtual machine that opens a separate window on the desktop display. It's preferable to manage the machines from a single environment, conveniently.

That brings us to the need to create a local cloud on the desktop. OpenStack is a popular, open source option and Red Hat offers an open source distribution (<http://openstack.rhdat.com>).

The RDO distribution of OpenStack will be included in the repositories of Fedora 21. You can add an additional repository for installing it on Fedora 20.

A bare bones cloud image is available from Fedora's download site. You can also build your own spin using or

expanding the kick-start for a cloud image, *fedora-x86_64-cloud.ks*, a part of the *fedora-kickstarts* package.

The plan

The process of exploring Big Data on a desktop needs to be broken up into smaller steps, with each step built on top of the previous step. Hopefully, it will run reasonably well on a quad-core desktop with 8 GB RAM to give you an understanding of the additional technology and the programming involved.

The current exploration will be on Fedora because my desktop is Fedora 20. Fedora offers an OpenStack distribution and it will be difficult to experiment on multiple distributions simultaneously.

The first step will be to create a local cloud.

Installing OpenStack

You could use a virtual machine to minimise the risk to your desktop environment. A useful utility is *appliance-creator*, which is a part of the *appliance-tools* package. You can use the kick-start file *fedora-x86_64-cloud.ks*, with a couple of changes in *fedora-cloud-base.ks*, to allow signing in as the root user. By default, the image requires *cloud-init* to create an account 'fedora' and inject ssh credentials for password-less logging in (see <https://www.technovely.org/linux/running-cloud-images-locally.html> as an alternate solution). You need to increase the size of the disk and SELinux should be permissive or disabled.

```
timezone --utc Asia/Kolkata
selinux --disabled
#rootpw --lock --iscrypted locked
rootpw some-password
part / --size 8192 --fstype ext4
#passwd -l root
```

You will need to make sure that the virtualisation packages are installed (https://fedoraproject.org/wiki/Getting_started_with_virtualization). Just do the following:

```
# yum install @virtualization
```